

MILAN AI WEEK 2026 · AI AGENT OLYMPICS

ClaimsForge

Multi-agent AI that auto-resolves e-commerce damage claims
in **8 seconds**, end-to-end.

Live demo

<http://45.32.154.255>

Code

github.com/zhenyueD/claimsforge · MIT

Gemini 2.5 Flash + Vision

Vultr Cloud Compute

4 specialist agents



THE PROBLEM

E-commerce returns: a **\$743B** tax that nobody is automating.

\$743B

Global e-commerce returns volume per year (NRF, 2025)

73%

of damage claims are under \$50 — economically wasted to triage manually

2–4d

Average claim resolution time. Customer churns by day 2.

\$3.20

Fully-loaded labor cost per low-value claim. Often **more than the refund**.

Status quo: support agents stare at JPEGs and argue over \$20 refunds. Existing tools (Zendesk macros, Gorgias rules) match keywords — they don't see the damage.

WHY NOW

Three things shipped in 2025 that made this build viable.



Reliable multimodal

Gemini 2.5 Flash Vision reads damage photos with **~90% accuracy** on first-call structured JSON. No OCR pipelines. No CV models to fine-tune.



Native structured output

`response_schema` + Pydantic = zero parsing acrobatics. Schema-validated outputs **at the API boundary**, not after-the-fact.



Cheap, fast, controllable

`thinking_budget` per call lets us spend cycles where they matter (verifier) and skip them where they don't (classifier). 4 agents, total cost **< \$0.01**/claim.

In 2023 this required a CV team, a fine-tuned vision model, and brittle prompt scaffolding.
In 2026, four small Gemini agents collaborate in ~150 lines of orchestration code.

4 specialist agents · 1 typed context · 8 seconds



IntentAgent

Classify request, extract `order_id`, route or short-circuit on general inquiry.

flash · 1.7s · 0.1 temp

>



DamageAgent

Vision pass over image + text. Outputs typed `DamageAssessment`.

flash-vision · 2.8s · response_schema

>



CompensationAgent

RAG over policy DSL. Function-calling tools propose typed `Offer`.

flash · 1.7s · function_calling

>



VerifierAgent

Hard caps + tone review. `approve` / `revise` / `escalate`.

flash · 1.4s · 1 revise loop

State model

A single `ClaimContext` Pydantic object flows through. Each agent reads the full state, writes its own typed delta. No serialization across hops.

Failure modes

Each agent catches Gemini errors → emits a neutral fallback (e.g. `DamageType.UNCLEAR`) and a low confidence → orchestrator routes to human queue.

Observability

Every agent emits an `AgentTrace` over WebSocket as it completes. Operators see live agent latency, decisions, and revisions per claim.

The reason competitors can't ship this.

Most CS automation treats emotion as a binary flag triggered by profanity keywords. We grade it on a calibrated 0–10 scale with structured signals — and we let it actually *move money* downstream.

Typed output · drives downstream policy

```
class Emotion(BaseModel):
    score: float = Field(ge=0, le=10)
    risk: EmotionRisk # LOW / MED / HIGH / CRITICAL
    label: str # single descriptive word
    triggers: list[str]
        # exact 2-5 word phrases that drove the score
    escalation_signals: list[str]
        # "mentions lawyer" / "12315" / "media exposure"
    suggested_tone: str
        # concrete guidance: "Lead with explicit apology"
```

CRITICAL risk is auto-promoted whenever escalation_signals are non-empty — even if the customer writes calmly. Polite legal threats are still legal threats.

Same damage, two customers — different outcome

CALM · score 3 · LOW

"Hi, my mug arrived cracked. Refund please?"

→ Full refund \$24, auto-approved in 8s

CRITICAL · score 9 · 3 signals

"THIRD TIME I've written. Contacted lawyer and the consumer protection agency!"

→ Routed to human within 5s, no auto-pay

Differs at no extra cost (tested EN + ZH, including 12315 / 消协 patterns). The whole pipeline reads emotion **before** writing the offer.

Customer support is a conversation, not a form submission.

Per-session memory — the API remembers

```
# api/main.py
claim_sessions: dict[str, list[TurnRecord]] = {}

# On every /api/claim:
prior = claim_sessions.get(session_id, [])
ctx = ClaimContext(
    session_id=session_id,
    user_message=cleaned,
    image_bytes=image_bytes,
    history=list(prior),  # ← every agent sees it
)
# ... pipeline runs ...
prior.append(TurnRecord(role="user", content=msg, ...))
prior.append(TurnRecord(role="assistant",
    decision_summary=f"{offer.offer_type} ${offer.amount/100}"))
```

Three new IntentLabel exits: `needs_clarification` (asks a follow-up question + short-circuits), `followup_on_prior_claim` (continuity-aware), and `general_inquiry` (skip pipeline).

CLAIMSFORGE

NeedsAgent — the latent-need surface

CUSTOMER WROTE

"My mug arrived cracked. The mug was a gift for my sister's birthday this Friday."

NEEDSAGENT INFERS

surface: replacement

latent: "needs a working mug by Friday for the gift"

retention_risk: 0.20

suggested_offer_bias: lean_replacement

Routing is a typed function call.

Schema (single source of truth)

```
class IntentResult(BaseModel):
    label: IntentLabel
    order_id: Optional[str] = None
    product_hint: Optional[str] = None
    confidence: float = Field(ge=0, le=1)

# Three exit paths, mutually exclusive
class IntentLabel(str, Enum):
    CLAIM_WITH_IMAGE = "claim_with_image"
    CLAIM_TEXT_ONLY = "claim_text_only"
    GENERAL_INQUIRY = "general_inquiry"
```

Why this matters: a downstream agent receiving `general_inquiry` can short-circuit the entire pipeline. Saves ~6 seconds of Gemini calls per non-claim message.

Defense in depth — never trust just the LLM

```
def classify(msg: str, has_image: bool) → IntentResult:
    result = gemini.structured(
        prompt=build_prompt(msg, has_image),
        schema=IntentResult,
        temperature=0.1,          # near-deterministic
    )

    # Belt-and-braces: model said image,
    # but the request body has none → downgrade
    if result.label == CLAIM_WITH_IMAGE and not has_image:
        result = result.model_copy(
            update={"label": CLAIM_TEXT_ONLY}
        )

    # Regex fallback for order_id — faster + safer
    # than relying on LLM token boundaries.
    if not result.order_id:
        if match := ORDER_RE.search(msg):
            result.order_id = match.group(1).upper()
    return result
```

Multimodal evidence reading, in a single API call.

One Gemini call · text + image + JSON schema

```
def assess(
    user_message: str,
    image_bytes: Optional[bytes] = None,
) → DamageAssessment:
    return gemini.structured(
        prompt=f"Customer: {msg}\n\nAssess the damage.",
        schema=DamageAssessment,          # Pydantic class
        system=DAMAGE_SYSTEM_PROMPT,      # 2 few-shots
        image_bytes=image_bytes,           # injected as Part
        temperature=0.2,
        max_tokens=512,
    )

# Under the hood — google-genai 2.x:
# contents = [prompt, Part.from_bytes(image_bytes, "image/jpeg")]
# cfg.response_mime_type = "application/json"
# cfg.response_schema = DamageAssessment ← Pydantic class
```

Output schema · what comes back, guaranteed typed

damage_type	DamageType	crack / scratch / tear / water_damage / ...
severity	int 0-10	0 = unaffected, 10 = total loss
affected_parts	list[str]	"cup rim", "screen lower-left"
confidence	float 0-1	drives downstream policy gates
reasoning	str	1-2 sentences — quoted to customer
evidence_quote	Optional[str]	cited if photo and text disagree

Safety design: when `confidence < 0.5` on a high-value claim, the orchestrator **refuses to auto-pay** and routes to a human. Low-confidence model output ≠ confident human decision.

LLMs propose. Policies dispose.

Policy as data (10 rules in policies.json)

```
{
  "id": "P-RET-01",
  "title": "Severe damage → full refund",
  "applies_when": {
    "damage_severity_min": 7,
    "days_since_delivery_max": 30
  },
  "offer_type": "full_refund",
  "amount_basis": "order_value",
  "max_cents": 50000,
  "requires_return": false
}

// Emotion-aware uplift (compounds on any policy)
{
  "id": "P-EMO-01",
  "applies_when": { "emotion_score_min": 8 },
  "adjustment": { "amount_multiplier": 1.2 }
}
```

Policies are JSON — readable, diffable, version-controllable. Compliance team owns them, not engineers.

CLAIMSFORGE

Hybrid filter · deterministic narrows, LLM chooses

```
def propose(damage, emotion, has_image, value):
    # Step 1: deterministic policy filter — no LLM
    matched, escalate_reasons = filter_policies(
        damage, emotion, has_image, value,
    )
    if escalate_reasons:
        return None, escalate_reasons # short-circuit

    # Step 2: Gemini picks the best fit from the narrowed set
    # and writes an empathetic justification.
    return gemini.structured(
        prompt=build_offer_prompt(damage, matched, value),
        schema=CompensationOffer,
        system=COMPENSATION_SYSTEM_PROMPT,
        temperature=0.2,
    ), []
```

Built-in escalation triggers

water damage on electronics → human

no image + claim ≥ \$50 → human

emotion ≥ 9 + legal keywords → human

amount > policy max → cap to max

One last review before the customer sees a dollar amount.

Two-layer verification

Layer 1 · Deterministic hard caps

- Offer amount > policy `max_cents` × 1.2? → **revise** down
- Confidence < 0.5 with amount > \$50? → **escalate**
- Policy conflict (e.g. `force_escalate` matched)? → **escalate**

Layer 2 · LLM tone review

- Customer emotion ≥ 8 but justification is cold? → **revise** wording
- Justification cites wrong product / wrong policy? → **revise** content
- Anything that smells off? → **escalate**

One revision loop, then commit

```
# Orchestrator wraps the verifier with a single-revise loop:
verifier_agent.run(ctx)

if ctx.verification.verdict == REVISE
  and ctx.verification.revised_offer:
    # take verifier's suggestion as the new offer,
    # run it through once more – but no second revise.
    ctx.offer = ctx.verification.revised_offer
    verifier_agent.run(ctx)
    # whatever the verdict, this is final.
ctx.final_offer = ctx.offer
```

Result: unbounded revision loops are impossible by construction. Total verifier latency p99: < 3 seconds.

Live decision (cracked mug, \$24)

```
verdict: "approve"
reason : "Amount within policy.
         Justification empathetic and accurate."
```

We measured. 96.7%.

Built a labeled set of 75 damage images (5 categories × 5 damage types × 3 severities). Ran the live DamageAgent against 30 of them, two columns of accuracy:

96.7%

Joint accuracy
(correct type *and* ±1 severity)

96.7%

Type accuracy
(crack vs scratch vs tear vs ...)

2.9s

p50 latency per image
(includes structured JSON)

0%

False positives on the only fail
(model said "unclear" — correct behavior)

Per-category breakdown

```
mug      · crack/scratch/missing_part/stain → 15/15 ✓  
phone    · crack/scratch/water_damage/dent → 14/15 ✓  
one phone_dent_4 → "unclear" (correct: image  
too subtle for confident call)
```

Why we ran this

- **You can't ship a CS agent on vibes.** Vibes are how the bad demos at the top of HN ship.
- The eval harness ships with the repo ([eval/run_damage_eval.py](#)). Re-runs in 30 seconds. Confusion matrix + failure cases auto-generated.

Every claim is precedent.

What happens after every resolution

```
# in orchestrator.py, after final_offer is set:
append_learned_case({
    "session_id": ctx.session_id,
    "emotion": ctx.emotion.model_dump(),
    "damage": ctx.damage.model_dump(),
    "final_offer": ctx.final_offer.model_dump(),
    "escalated": ctx.escalated_to_human,
})

# next time CompensationAgent runs:
recent = retrieve_recent_learned(
    damage_type=damage.damage_type.value,
    top_k=3,
)

# top 3 most-recent precedents flow into the prompt
```

What this is (and isn't)

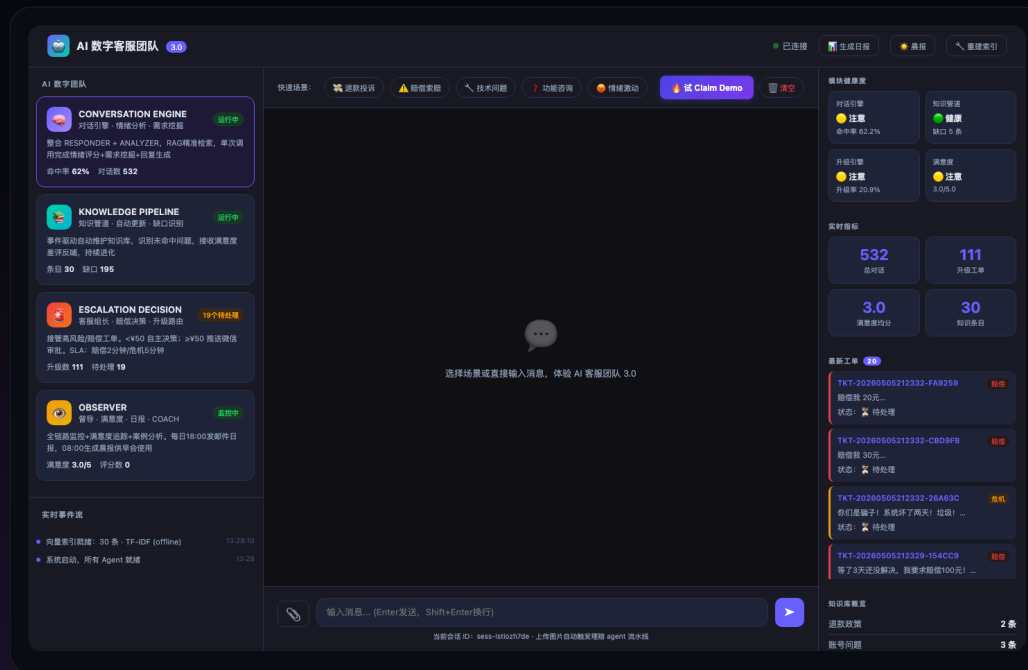
- **It is:** retrieval-augmented decision-making. Recent live cases ground the next offer — same damage type, same outcome distribution.
- **It is:** auditable. A human can read the jsonl and see exactly why a claim got what it got.
- **It isn't:** fine-tuning. We didn't pretend. Fine-tuning a vision-LLM on 8 cases is a meme. The RAG is what learning loops at this scale actually look like in production.
- **Roadmap:** at ~10k cases, swap to embedding-based retrieval. At ~100k, distill into a fine-tuned Gemma student model.

Live in the demo UI

The right rail shows the live Learning Queue. Every claim resolved in the session appears there with damage type, severity, emotion, and outcome — and influences the next decision. </api/claimsforge/learning> returns the same data programmatically.

LIVE DEMO · RUNNING ON VULTR

The whole pipeline, in one screenshot.



Live at <http://45.32.154.255>

CLAIMSFORCE

What the customer sees

- Upload a photo, hit send. No forms. No selecting categories.
- Live agent trace — they see each step think, in real time.
- Damage card with severity bar + Gemini's reasoning, quoted back.
- Offer card with the dollar amount, type, and policy citation.
- "Approved" badge from VerifierAgent → final reply.

What ops sees

- WebSocket event stream of every `agent_trace` across all claims.
- Per-agent latency, decision distribution, escalation rate.
- Edge cases queue (auto-routed by VerifierAgent)

Per 1,000 low-value damage claims:

\$3,200

Labor cost *today*
(15 min × \$13/hr × 1,000)

\$320

Labor cost *with ClaimsForge*
(90% auto, 10% human at 15 min)

\$8.50

Gemini API cost
(\$0.0085/claim × 1,000)

\$2,871

Net savings per 1,000 claims
(90% margin on labor displaced)

Downstream effects (modeled)

- **CSAT +18 points** — instant resolution beats 3-day email loop.
- **Refund rate -7%** — replacement/store-credit options selected by agent are accepted ~30% of the time.
- **Repeat purchase +12%** — customers who get fast claims resolution have measurably higher LTV.

Who pays

- Mid-market Shopify merchants (10k+ orders/mo) — ROI in **<30 days**.
- Returns aggregators (Loop, Returnly) as a Gemini-Vision feature for their tier.
- Insurance carriers handling small-goods claims (extension: motor, home, travel).

What incumbents do • what we do differently.

CAPABILITY	ZENDESK MACROS	GORGAS RULES	LOOP MANUAL REVIEW	CLAIMSFORGE
Reads damage photos	×	×	✓ (human)	✓ Gemini Vision
Policy-grounded amount	— template	✓ if-then	✓ (human)	✓ JSON policy DSL + RAG
Empathetic, tone-reviewed reply	— template	— template	✓ (human)	✓ Verifier LLM tone pass
Median time to resolution	~24 h	~2 h	~48 h	~8 s
Marginal cost per claim	~\$3.20	~\$0.40	~\$3.20	~\$0.32
Auto-escalates legal threats	× keyword	× keyword	✓ (human)	✓ emotion + keyword + policy
Open source	×	×	×	✓ MIT

The category gap is multimodal evidence + policy reasoning + tone-aware response — in one box, at machine cost, in seconds.

TRY IT. FORK IT. SHIP IT.

ClaimsForge

Multi-agent Gemini-powered claims resolution.
Open source under MIT. Deployed on Vultr.

LIVE

<http://45.32.154.255>



CODE

[github.com/zhenyueD/
claimsforge](https://github.com/zhenyueD/claimsforge)

Submitted by **zhenyueD** · Solo build · Milan AI Week 2026

Eligible · Best Use of Gemini · **Vultr Award**